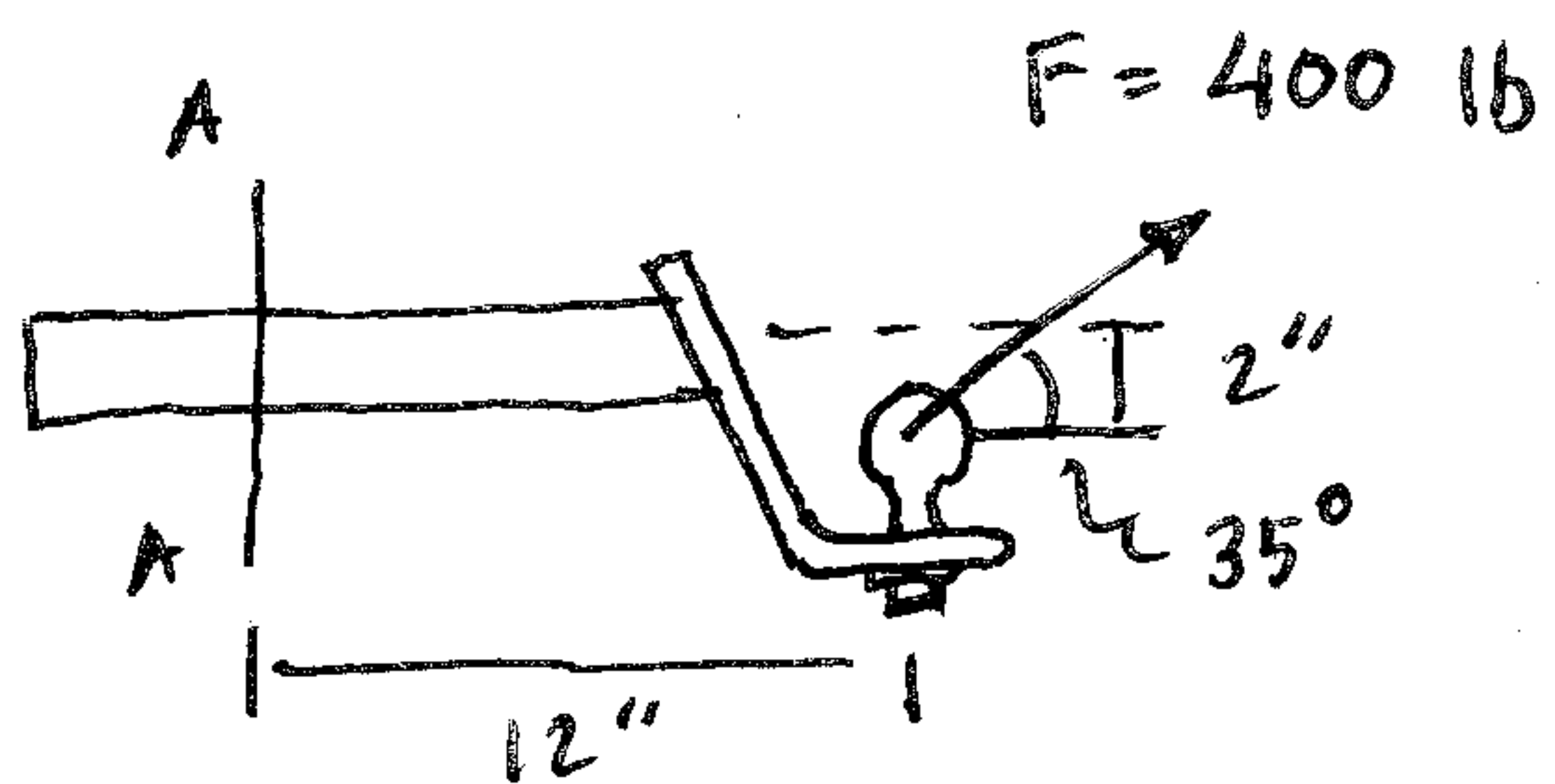
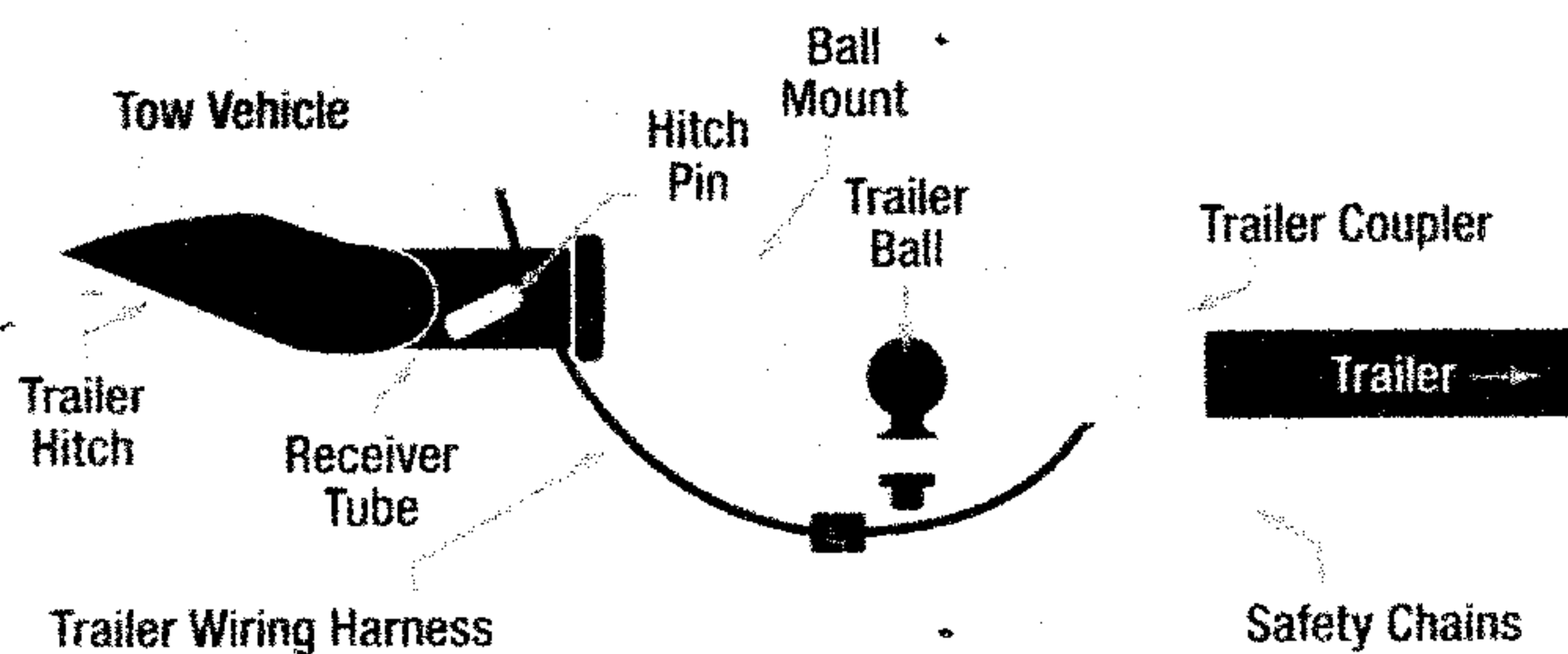


Internal Forces Trailer Hitch Problem



Determine the internal forces at section AA.

$$12'' \uparrow 400 \sin 35 = 229.4 \text{ lb}$$

$$2'' \rightarrow 400 \cos 35 = 327.7 \text{ lb}$$

$$A = 327.7 \text{ lb}$$

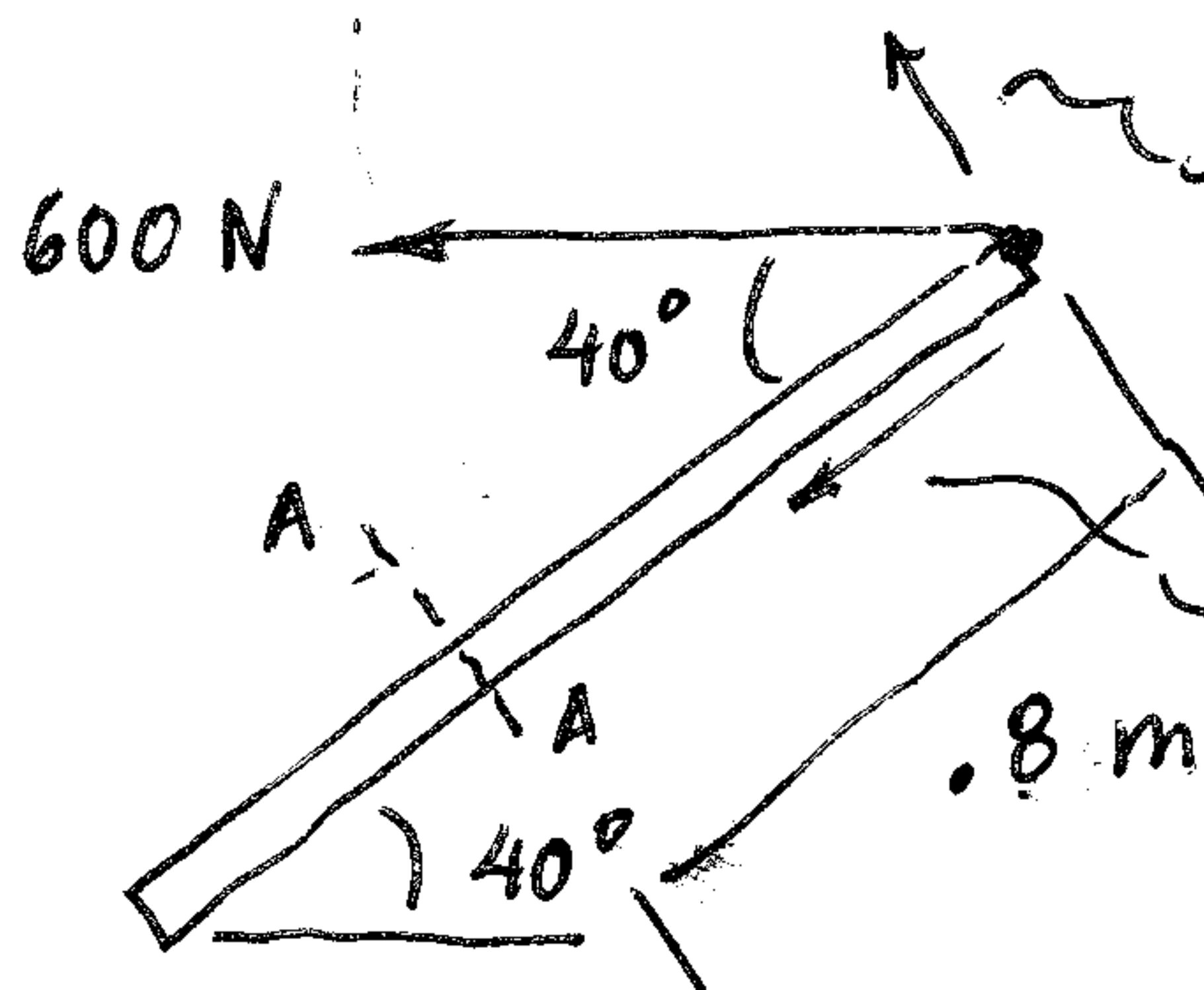
$$V = 229.4 \text{ lb}$$

$$M = 327.7(2'') + 229.4(12'')$$

$$= 3.41 \cdot 10^3 \text{ lb-in}$$

Angled Member Problem

Determine the internal forces at section AA:



$$600 \sin 40 = 385.7 \text{ N}$$

$$600 \cos 40 = 459.6 \text{ N}$$

$$A = 459.6 \text{ N}$$

$$V = 385.7 \text{ N}$$

$$M = 385.7(0.8) = 308.6 \text{ N}\cdot\text{m}$$

Note: This kind of problem almost always asks for the magnitudes of the axial, shear, and bending moments.